

MULTIPLE HYPOTHESES AND CLUSTERING FOR NORMAL MEANS - A BAYESIAN ANALYSIS

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Summary: For the one-way analysis of variance model, multiple precise hypotheses are defined regarding the means. These hypotheses describe different possible models or clusterings of the means in lower-dimensional subspaces of the whole parameter space. The means are assumed to have been generated by exchangeable normal prior distributions. A positive prior mass is placed on each of these subspaces and the resulting posterior probabilities are derived as functions of one unknown parameter, the variance ratio. Two applications are given, one concerning the determination of a possible change-point in the prevalence rate of a congenital malformation.

1. INTRODUCTION

Consider N random samples, each of size n , from normal populations. The samples are independent conditional on the population means and common variance:

$$(1.1) \quad X_{ij} | m_j, \sigma^2 \sim N(m_j, \sigma^2) \quad i = 1, \dots, N; j = 1, \dots, n.$$

This is the classical one-way analysis of variance model and has been extensively analysed within a Bayesian framework. In particular, see Lindley and Smith (1972) for the more general linear model, and Box and Tiao (1973) for a discussion of fixed effect and random effect models. The results

of these analyses are given in terms of joint and marginal posterior density functions of the parameters.

Our purpose is slightly different. By defining some precise hypotheses regarding the population means and by placing a positive prior mass on the parameter space under each of the hypotheses, posterior probabilities for all hypotheses can be obtained. It can then be determined if the hypothesis of equality of means is reasonable, and if not, which of the multiple alternatives is the most likely. These alternatives define different models, or clusterings, into $r < N$ populations. This enables us to do multiple comparisons directly.

A sample-theoretic approach to clustering in one-way analysis of variance is given in Scott and Knott (1974). This is a testing procedure, derived under the assumption of homogeneity of means, and the results can therefore differ substantially from those obtained using the Bayesian analysis of this paper. A number of other sample-theoretic clustering algorithms are available in the literature, see, for example, Anderberg (1973) and Hartigan (1975). A method closer to the pure Bayesian approach is the "quasi-Bayes" procedure proposed by Geisser (1974), Geisser and Eddy (1979) for model selection. They used predictive sample re-use to determine which of a number of models yields the best predictions for future observations.

A Bayesian approach to cluster analysis is given by Binder (1978). It is not the one-way analysis of variance model considered here, but is concerned with the grouping of a number of multi-dimensional observations into r groups, where r is fixed. The approach is decision-theoretic and the criterion is to minimize expected loss. A Bayes rule for multiple comparisons is proposed by Duncan (1965) and Waller and Duncan (1969). It is a multiple decision theory model and is not concerned with clustering.

2. MULTIPLE HYPOTHESES

For the model in (1.1) a number of hypotheses, one for each possible clustering of population means, can be defined. Let H_{rk} denote the

hypothesis that the data was generated by r means, and the index k denotes a particular clustering into r groups, where $r = 1, \dots, N$. For simplicity the k is an arbitrary index, in order to distinguish between different partitionings into the same number of groups. In an application the particular partitioning for a given k must be defined. So H_{11} denotes the hypothesis that all the means are equal, while H_{N1} denotes the hypothesis that all means are different.

Theoretically, all possible clusterings of N samples into r groups can be included in the set of hypotheses, but because of the combinatorial explosion of possible clusters when $N \geq 5$, it is in practice convenient to consider only those that are considered feasible or are of specific interest.

The hypothesis H_{rk} defines a parameter space in r dimensions, a subspace of the complete N -dimensional parameter space. Under H_{rk} we assume that

$$(2.1) \quad M_i | H_{rk} \sim N(\mu_i, \gamma \sigma^2), \quad i = 1, \dots, r$$

independently, and this prior is exchangeable between hypotheses. This means that (2.1) holds for all r and k . We assume, for the moment, that γ is known.

In the absence of any information concerning the hyper-parameters μ and σ^2 , the usual Bayesian analysis resorts to improper priors. This works well in estimation problems but is not always appropriate in tests of significance since the posterior probability can be extremely sensitive to the improper prior used under a particular hypothesis (DeGroot (1982)). However, if the improper priors on some hyperparameters are exchangeable between hypotheses, the resulting posterior probability is quite robust with respect to the exact form of the priors on these hyperparameters. To show this, we will start with proper second-stage priors on μ and σ^2 , i.e.

$$(2.2) \quad \mu | \sigma^2 \sim N(\phi, \tau \sigma^2) \text{ and } \sigma^{-2} \sim \text{Gamma}(\alpha, \beta)$$

with parameters ϕ, τ, α and β assumed known.

Let p_{rk} be the prior probability for H_{rk} , where $\sum_r \sum_k p_{rk} = 1$. Under

H_{rk} the mass p_{rk} is then distributed over the $(r+2)$ -dimensional space, $\underline{m}_r \in R^r$, $\mu \in R$, $\sigma^2 > 0$, according to the joint prior distribution;

$$(2.3) \quad \Pi_{rk}(\underline{m}_r, \mu, \sigma^2) = \prod_{i=1}^r n(m_i | \mu, \gamma \sigma^2) h(\mu) g(\sigma^2),$$

where

$$\underline{m}_r = (m_1, \dots, m_r)'$$

The likelihood function is given by

$$(2.4) \quad L(H_{rk}) = \prod_{i=1}^r \prod_{j=1}^{q_{ik}} n(x_{ij} | m_i, \sigma^2),$$

where q_{ik} denotes the size of the i -th sample when grouped according to hypothesis H_{rk} , and $\sum_{i=1}^r q_{ik} = Nn$ for all k .

According to Bayes' theorem, the posterior probability for H_{rk} , denoted by λ_{rk} , multiplied by the joint posterior under H_{rk} , is given by (X represents all the data):

$$(2.5) \quad \lambda_{rk} \Pi_{rk}(\underline{m}_r, \mu, \sigma^2 | X) = \frac{p_{rk} \Pi_{rk}(\underline{m}_r, \mu, \sigma^2) L(H_{rk})}{\sum_s \sum_t p_{st} \int \dots \int \Pi_{st}(\underline{m}_s, \mu, \sigma^2) L(H_{st}) d\underline{m}_s d\mu d\sigma^2}$$

where the summations over t and s include all the models being considered.

The rk -th term in the denominator is given by

$$(2.6) \quad D_{rk} = p_{rk} \int_0^\infty \int_{-\infty}^\infty \dots \int_{-\infty}^\infty \prod_{i=1}^r \left[n(m_i | \mu, \gamma \sigma^2) \prod_{j=1}^{q_{ik}} n(x_{ij} | m_i, \sigma^2) \right] \cdot n(\mu | \phi, \tau \sigma^2) \frac{\beta^2}{\Gamma(\alpha)} (\sigma^2)^{-(\alpha+1)} e^{-\frac{\beta}{\sigma^2}} d\underline{m}_r d\mu d\sigma^2$$

$$\begin{aligned}
&= p_{rk} (2\pi)^{-\frac{1}{2}(Nn+r+1)} \gamma^{-\frac{1}{2}} \tau^{-\frac{1}{2}} \frac{\beta^2}{\Gamma(\alpha)} \\
&\quad \cdot \int_0^\infty (\sigma^2)^{-\frac{1}{2}(Nn+r+2\alpha+3)} \exp\left(-\frac{\beta}{\sigma^2}\right) \\
&\quad \cdot \int_{-\infty}^\infty \dots \int_{-\infty}^\infty \exp\left\{-\frac{1}{2\sigma^2} \sum_{i=1}^r \sum_{j=1}^{q_{ik}} (x_{ij} - m_i)^2\right\} \\
&\quad \cdot \exp\left\{-\frac{1}{2\sigma^2} \left[\frac{1}{\gamma} \sum_{i=1}^r (m_i - \mu)^2 + \frac{1}{\tau} (\mu - \phi)^2\right]\right\} d\mu d\mathbf{m}_r d\sigma^2 \\
&= p_{rk} (2\pi)^{-\frac{1}{2}(Nn+1)} \gamma^{-\frac{1}{2}} \tau^{-\frac{1}{2}} \left(\frac{1}{\gamma} + \frac{1}{\tau}\right)^{-\frac{1}{2}} \tau^{-\frac{1}{2}} \int_0^\infty (\sigma^2)^{-\frac{1}{2}(Nn+r+2\alpha+2)} \\
&\quad \cdot \int_{-\infty}^\infty \dots \int_{-\infty}^\infty \exp\left\{-\frac{1}{2\sigma^2} (\mathbf{m}_r - \mathbf{R}_{rk}^{-1} \mathbf{P}_{rk})' \mathbf{R}_{rk} (\mathbf{m}_r - \mathbf{R}_{rk}^{-1} \mathbf{P}_{rk})\right\} \\
&\quad \cdot \exp\left\{-\frac{1}{2\sigma^2} \mathbf{u}_{rk}' \mathbf{u}_{rk}\right\} d\mathbf{m}_r d\sigma^2,
\end{aligned}$$

where

$$(2.7) \quad \mathbf{x}_i = (x_{i1}, x_{i2}, \dots, x_{in})',$$

$$\bar{\mathbf{x}}_{rk} = (\bar{x}_{1k}, \dots, \bar{x}_{rk})', \quad \bar{\mathbf{x}}_{rk} = \frac{1}{q_{ik}} \sum_{j=1}^{q_{ik}} \mathbf{x}_{ij},$$

$$\mathbf{Q}_{rk}(r \times r) = \text{diag}(q_{1k}, \dots, q_{rk}),$$

$$\mathbf{R}_{rk}(r \times r) = \frac{1}{\gamma} (\gamma \mathbf{Q}_{rk} + \mathbf{I}_r - \frac{\tau}{\tau r + 1} \mathbf{e} \mathbf{e}'),$$

$$\mathbf{P}_{rk}(r \times 1) = \mathbf{Q}_{rk} \bar{\mathbf{x}}_{rk} + \frac{\phi}{\tau r + 1} \mathbf{e},$$

$$\mathbf{u}_{rk} = \sum_{i=1}^N \mathbf{x}_i' \mathbf{x}_i + \frac{\phi^2}{\tau} \left(1 - \frac{\gamma}{\tau r + 1}\right) - \mathbf{P}_{rk}' \mathbf{R}_{rk}^{-1} \mathbf{P}_{rk} + 2\beta,$$

$$\mathbf{e}' = (1, \dots, 1).$$

Then

$$(2.8) \quad D_{rk} = p_{rk} \pi^{-\frac{1}{2}(Nn+r)} \gamma^{-\frac{1}{2}r} \tau^{-\frac{1}{2}} \left(\frac{1}{\gamma} + \frac{1}{\tau} \right)^{-\frac{1}{2}} \Gamma\left[\frac{1}{2}(Nn+r+2\alpha-1)\right] \\ \cdot \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} [u_{rk} + (\underline{m}_r - \underline{\theta}_{rk})' R_{rk} (\underline{m}_r - \underline{\theta}_{rk})]^{-\frac{1}{2} [Nn+r+2\alpha-1]} d\underline{m}_r$$

where

$$(2.9) \quad \underline{\theta}_{rk} = R_{rk}^{-1} \underline{p}_{rk}.$$

So

$$(2.10) \quad D_{rk} = p_{rk} \pi^{\frac{1}{2}Nn} \gamma^{-\frac{1}{2}r} \tau^{-\frac{1}{2}} \left(\frac{1}{\gamma} + \frac{1}{\tau} \right)^{-\frac{1}{2}} u_{rk}^{-\frac{1}{2} [Nn+2\alpha-1]} |R_{rk}|^{-\frac{1}{2}} \Gamma\left[\frac{1}{2}(Nn+2\alpha-1)\right],$$

and the posterior probability for hypothesis H_{rk} is given by

$$\lambda_{rk} = \frac{D_{rk}}{\sum_s \sum_t D_{st}}$$

The joint posterior distribution under H_{rk} is proportional to $\Pi_{rk}(\underline{m}_r, \mu, \sigma^2) L(H_{rk})$ and is given, after some algebraic manipulation, by

$$(2.11) \quad \Pi_{rk}(\underline{m}_r, \mu, \sigma^2 | X) = \prod_{i=1}^r f_{rk}(m_i | \mu, \sigma^2, X) h_{rk}(\mu | \sigma^2, X) g_{rk}(\sigma^2 | X)$$

where

$$f_{rk}(m_i | \mu, \sigma^2, X) = n \left[m_i \left| \frac{\gamma q_{ik} \bar{x}_{ik} + \mu}{\gamma q_{ik} + 1} ; \frac{\gamma \sigma^2}{\gamma q_{ik} + 1} \right| \right],$$

$$h_{rk}(\mu | \sigma^2, X) = n \left[\mu | \hat{\mu}_{rk}; \sigma^2 \left[\sum_{i=1}^r \frac{q_{ik}}{\gamma q_{ik} + 1} + \frac{1}{\tau} \right]^{-1} \right],$$

$$\hat{\mu}_{rk} = \left[\sum_{i=1}^r \frac{q_{ik} \bar{x}_{ik}}{\gamma q_{ik} + 1} + \frac{\phi}{\tau} \right] \left[\sum_{i=1}^r \frac{q_{ik}}{\gamma q_{ik} + 1} + \frac{1}{\tau} \right]^{-1}$$

and $g_{rk}(\sigma^2 | X)$ is an inverted gamma distribution with parameters u_{rk} and $Nn + 2\alpha - 1$.

The marginal posterior distribution of $\underline{m}_r$ under H_{rk} is a multivariate t-distribution with $Nn - 1$ degrees of freedom,

$$(2.12) \quad E[\underline{M}_r | H_{rk}, X] = \underline{\theta}_{rk},$$

$$\text{Cov}[\underline{M}_r, \underline{M}_r' | H_{rk}, X] = \frac{u_{rk}}{Nn + 2\alpha - 3} R_{rk}^{-1}.$$

It can be seen from the preceding formulae that if the second-stage priors are diffused so that τ is relatively large and α and β are relatively small in comparison with the other quantities, then their effect on the posterior probabilities would be minimal.

If we let $\tau \rightarrow \infty$ and $\alpha, \beta \rightarrow 0$, then the posterior probability for H_{rk} is

$$(2.13) \quad \lambda_{rk} = \frac{T_{rk}}{\sum_s \sum_t T_{st}}$$

where the summation is over all hypotheses, and

$$(2.14) \quad T_{st} = p_{st} s^{-\frac{1}{2}} \gamma^{-\frac{s-1}{2}} u_{st}^* \frac{Nn-1}{2} |R_{st}^*|^{-\frac{1}{2}}$$

with

$$(2.15) \quad R_{rk}^* = \frac{1}{\gamma\tau} (\gamma\tau Q + rI_r - ee'),$$

$$u_{rk}^* = \sum_{i=1}^N \underline{x}_i' \underline{x} - \bar{\underline{x}}_{rk}' Q_{rk} R_{rk}^{-1} Q_{rk} \bar{\underline{x}}_{rk}.$$

3. DISCUSSION

The behaviour of the posterior probability γ_{rk}

The posterior probability γ_{rk} given in (2.13) is a function of the variance ratio γ , which would generally be unknown. The probabilities can be examined over a range of values for γ , particularly in the neighbourhood of some estimated value for γ .

When $\gamma \rightarrow 0$, then $\gamma_{rk} \rightarrow p_{rk}$, the prior probability for H_{rk} , and the evidence from the data is ignored. On the other hand, when $\gamma \rightarrow \infty$, so that the priors become very diffused, all the posterior probability concentrates on the null hypothesis and $\gamma_{11} \rightarrow 1$ while $\gamma_{rk} \rightarrow 0$ for all k and $r \neq 1$. This

behaviour is sometimes called "Jeffrey's paradox" (or "Lindley's paradox") (Lindley (1957), Shafer (1982)), since the effect of the diffused prior in testing seems to be contradictory to the effect in estimation. The reason for the apparent paradox is that a positive mass is placed on a subspace of lower dimension than the whole parameter space.

If H_{11} is not included in the set of hypotheses, all the posterior probability will be distributed among the set of hypotheses of lowest dimension as $\gamma \rightarrow \infty$, so that

$$(3.1) \quad \lim_{\gamma \rightarrow \infty} \lambda_{sk} = \left[1 + \frac{\sum_{t \neq k} p_{st} S_{st}^{\frac{Nn-1}{2}} \left[\prod_{i=1}^s q_{it} \right]^{-\frac{1}{2}}}{p_{sk} S_{sk}^{\frac{Nn-1}{2}} \left[\prod_{i=1}^s q_{ik} \right]^{-\frac{1}{2}}} \right]^{-1}$$

where s is the lowest dimension considered,

$$S_{sk} = \sum_i \sum_j q_{ijk} (x_{ij} - \bar{x}_{ik})^2,$$

and $\lim_{\gamma \rightarrow \infty} \lambda_{tk} = 0$ for all $t > s$.

An empirical Bayes estimator for γ , based on the conditional and unconditional moments of the data, and derived under the particular hypothesis, H_{rk} , is given by

$$(3.2) \quad \hat{\gamma}_{rk} = \left[(r-2) + \frac{r \sum_i q_{ik}^2}{N^2 n^2} \right]^{-1} \left[\frac{\sum_i (\bar{x}_{ik} - \bar{\bar{x}})^2}{\hat{\sigma}_{rk}^2} - \sum_i \frac{1}{q_{ik}} + \frac{1}{n} \right], \quad r \geq 2$$

where

$$(3.3) \quad \hat{\sigma}_{rk}^2 = \frac{1}{Nn} \sum_i \frac{q_{ik}}{q_{ik} - 1} \sum_j q_{ijk} (x_{ij} - \bar{x}_{ik})^2,$$

$$\bar{\bar{x}} = \frac{1}{Nn} \sum_i \sum_j x_{ij}.$$

This estimator is a function of the ratio of between - and within - groups variances. For example, when all groups are considered different with $r = N$ and $q_{ik} = n$ for all i , then (3.2) reduces to

$$\hat{\gamma}_{N1} = f - \frac{1}{n},$$

where f is the F-statistic:

$$f = \frac{N(n-1) \sum_{i=1}^N (\bar{x}_i - \bar{\bar{x}})^2}{(N-1) \sum_{i=1}^N \sum_{j=1}^n (x_{ij} - \bar{x}_i)^2}$$

Notice that the problem is undefined under the hypothesis of equality, H_{11} , and no estimator for γ is possible, but a single estimator for γ can, however, be defined as

$$(3.4) \quad \hat{\gamma} = \frac{\sum_{r \neq 1} \sum_k p_{rk} \hat{\gamma}_{rk}}{\sum_{r \neq 1} \sum_k p_{rk}}$$

The prior probability p_{rk}

If the data strongly favours one particular hypothesis, the assigned prior probabilities will have little effect on the posterior probabilities when γ moves away from zero. In other cases, however, the subjectively assigned prior probabilities can have significant effect on the posterior probabilities, and in the absence of specific prior information, the ideal is to assign "neutral" or impartial prior probabilities. The problem is representing impartiality.

If the first objective of the analysis is to determine whether the means can be considered equal, as in one-way analysis of variance, and the determination of the best clustering, given the means are not all equal, is a secondary objective, then the assignment of a prior mass of $\frac{1}{2}$ to the hypothesis of equality seems appropriate. The rest of the prior probability can be uniformly distributed among all the other hypotheses.

If, on the other hand, there is no reason to assume equality above any other hypothesis, and the purpose of the analysis is simply to find the most appropriate model, or clustering, then the prior probability can be uniformly distributed over all hypotheses.

The final posterior probabilities will of course depend on the construction of the parameter space, that is, the hypotheses included in the analysis. The posterior probability for the null hypothesis of homogeneity against a single alternative distributed over the whole parameter space can differ from the result obtained against multiple alternatives defined over subspaces. If the purpose of the analysis is to find the best model, this discrepancy is not so important, since the interest is in the comparative values of posterior probabilities. If the actual posterior probabilities are of interest for possible further analysis, more care should be taken in defining the relevant parameter space.

4. APPLICATIONS

Example 1

As a first example, consider the data on total Fe for four types of iron formation from the article "Origin of Precambrian Iron Formations" in *Econ. Geology*, as reproduced in Devore (1987, p. 378). (A = hematite, B = magnetite, C = carbonate, D = silicate). There are $n = 10$ observations for each type of iron formation and these are summarized in table 1.

Table 1. Total Fe for four types of iron formation

Type	A	B	C	D
Mean	33,84	29,95	26,08	24,69
Sum of squares	210,064	73,305	103,516	176,249

Our purpose is to determine which clustering of the types of formation has the highest posterior probability. As there is no special motivation for assuming homogeneity of means, the prior probability will be uniformly distributed over all possible clusterings. For $N = 4$ there are 15 possible clusterings, from H_{11} : all means equal, up to H_{41} : All means different. Each of these 15 hypotheses is assigned a prior probability of 0,0667. Obviously some of these hypotheses will have very small posterior probabilities, and in figure 1 only the four hypotheses with the highest posterior probabilities are represented as functions of the variance ratio γ . These four are:

$$H_{25}: AB|CD$$

$$H_{31}: A|B|CD$$

$$H_{32}: AB|C|D$$

$$H_{41}: A|B|C|D$$

The highest posterior probability is attained by H_{31} over the range $0,1 < \gamma < 15$. For $\gamma \geq 15$, H_{25} takes over as hypothesis with highest posterior probability. Of course, as $\gamma \rightarrow \infty$, H_{11} will eventually have the highest posterior probability as discussed in section 3. For reasonable values of γ , however, the posterior probability for H_{11} is virtually zero.

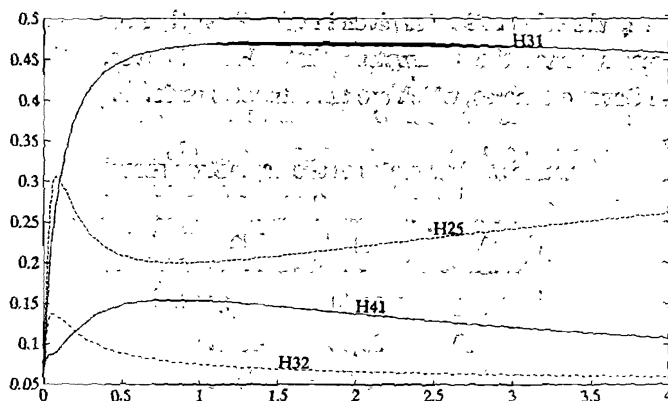


Figure 1. Posterior probabilities for four hypotheses

The empirical Bayes estimator of γ follows from (3.4) as $\hat{\gamma} = 0,622$ which does suggest that γ is well into the range for which H_{31} is our best model. The posterior probabilities of all 15 hypotheses at $\hat{\gamma}_3 = 0,622$ are given by:

$$\begin{aligned} \lambda_{11} &= 0,0001, \lambda_{21} = 0,0374, \lambda_{22} \approx 0, \lambda_{23} = 0,0001, \lambda_{24} = 0,0014, \\ \lambda_{25} &= 0,2032, \lambda_{26} = 0,0001, \lambda_{27} \approx 0, \lambda_{31} = 0,4564, \lambda_{32} = 0,0834, \\ \lambda_{33} &= 0,0005, \lambda_{34} = 0,0001, \lambda_{35} = 0,0525, \lambda_{36} = 0,0128, \lambda_{41} = 0,1520. \end{aligned}$$

The data supports only the four hypotheses shown in figure 1, in the sense that the Bayes factors,

$$B_{rk} = \frac{1 - p_{rk}}{p_{rk}} \frac{\lambda_{rk}}{1 - \lambda_{rk}}, \text{ are larger than one. They are}$$

$$B_{31} = 11,8, B_{25} = 3,6, B_{41} = 2,5, B_{32} = 1,3.$$

The posterior distribution of the cluster means is a mixture of t -distributions in different dimensions. Point estimates of the population means can be obtained as weighted averages of the posterior means under each of the 15 hypotheses with weights equal to their respective posterior probabilities. These estimates are denoted by $\hat{\theta}$ in table 2. For comparison, the other estimates given in table 2 are the sample means ($\bar{X}$), the estimates under the assumption that H_{31} is true, and the usual empirical Bayes estimates, which are the estimates under H_{41} .

Table 2. Estimates of the population means

	A	B	C	D
$\hat{\theta}$	32,76	30,14	26,04	25,62
H_{31}	33,50	29,93	25,57	25,57
H_{41}	33,36	29,83	26,32	25,05
$\bar{X}$	33,84	29,95	26,08	24,69

The sample means are shrunk towards their common mean to obtain the empirical Bayes estimates (H_{41}); while for θ , because of the relatively high posterior probabilities of H_{31} and H_{25} , the means of C and D are shrunk more towards each other. Under H_{31} these two means are equal, while there is less shrinkage effect on A and B.

For multiple comparisons, the posterior probability of equality of any two means (or more) can be obtained directly from the posterior probabilities of all hypotheses. It follows that

$$\begin{aligned} P[m_C = m_D | X] &= 0,6971, & P[m_B = m_D | X] &= 0,0505 \\ P[m_A = m_B | X] &= 0,2882, & P[m_A = m_C | X] &= 0,0021 \\ P[m_B = m_C | X] &= 0,0914, & P[m_A = m_D | X] &= 0,0003. \end{aligned}$$

One-way analysis of variance of this problem yields an observed F-value of 10,82, which is highly significant, while Tukey's procedure for multiple comparisons judged the last three of the above comparisons to be significantly different at a 0,01 significance level.

The Scott-Knott procedure for clustering (Scott and Knott (1974)) divides the data into the two groups of hypothesis H_{25} when the significance level is $\alpha = 0,01$, and into four groups when $\alpha = 0,05$.

Example 2 (Change-point testing)

A special case of the preceding theory is the problem of a change in mean in a sequence of normal random variables. Here there is a major simplification because of the natural order in which the observations appear, and substantial literature on the Bayesian analysis of change-point problems exists. For a review, see Broemeling and Tsurumi (1987), whose approach differs from the treatment in this paper mainly in the assumptions that all hyperparameters are known and that there exists exactly one change-point. Other papers dealing with this problem within a Bayesian framework are Chernoff and Zacks (1963), Sen and Srivastava (1973), Smith (1975), Broemeling (1977) and Vee Ming Ng (1990), but the approaches, assumptions and objectives differ from those in this paper.

In table 3 the data on observed prevalence of a congenital malformation (Hypospadias) in the Liverpool area, from Simpkin and Downham (1988) are given for the years 1960-1982. The proportion malformations, $\hat{p}_i$, were obtained from large samples, varying from 7 000 to 18 000.

Table 3. Proportion malformations

Year	60	61	62	63	64	65
$\hat{p}_i (\times 10^{-3})$	1,449	1,179	1,156	1,589	1,008	1,172

Year	66	67	68	69	70	71
$\hat{p}_i (\times 10^{-3})$	1,656	1,360	0,452	2,132	0,830	2,224

Year	72	73	74	75	76	77
$\hat{p}_i (\times 10^{-3})$	2,888	2,543	2,880	2,705	2,348	2,011

Year	78	79	80	81	82
$\hat{p}_i (\times 10^{-3})$	2,933	2,160	2,199	2,146	2,948

Simpkin and Downham (1988) were interested in testing the null hypothesis of a constant malformation rate over time,

$$H_0: p_i = p, \quad i = 1, \dots, N$$

against an alternative of a single change-point in rate,

$$H_1: p_i = \begin{cases} \alpha, & i = 1, \dots, k, \\ \beta, & i = k+1, \dots, N, \end{cases}$$

where k is unknown.

To apply the theory of section 2, let $x_i = 0,081 \arcsin \sqrt{\hat{p}_i} - 0,001561$. The transformed data is then approximately normally distributed, and the variances stabilized. The constants were chosen so that the mean and the variance of the data remain unaffected by the transformation. Then

$$X_i | m_i, \sigma^2 \sim N(m_i, \sigma^2),$$

and the null hypothesis is

$$H_0: m_1 = \dots = m_N = m_0,$$

against the $N - 1$ alternatives,

$$H_k: (m_1 = \dots = m_k) \neq (m_{k+1} = \dots = m_N), k = 1, \dots, N - 1.$$

For this problem it follows from (2.13) and (2.14) that the posterior probabilities are given by

$$(4.1) \quad \lambda_k = \frac{\sum_{j=0}^{N-1} p_j (2\gamma j(N-j) + N)^{-\frac{1}{2}} u_j^{\frac{N-1}{2}}}{p_k (2\gamma k(N-k) + N)^{-\frac{1}{2}} u_k^{\frac{N-1}{2}}}, \quad k = 0, \dots, N - 1,$$

where

$$u_k = \sum_{i=1}^N x_i^2 - (2k(N-k)\gamma + N)^{-1} [(k\bar{x}_k + (N-k)\bar{x}_{N-k})^2 + 2\gamma k(N-k)(k\bar{x}_k^2 + (N-k)\bar{x}_{N-k}^2)].$$

With a prior probability of $p_0 = \frac{1}{2}$ for H_0 and $p_k = \frac{1}{44}$, $k = 1, \dots, 22$, for the alternative hypotheses, the posterior probability λ_{11} quickly dominates all others. In figure 2, four of the posterior probabilities are shown as functions of γ , all others being practically zero. It can be seen that the data heavily favours a change-point between 1970 and 1971 while the probability of no change-point goes to almost zero.

Under the assumption that H_{11} is true, the empirical Bayes estimator of γ follows from (3.2) as $\hat{\gamma}_{11} = 5,0$, where $\lambda_{11} = 0,866$. The posterior

distribution of $\underline{m} = \begin{bmatrix} m_1 \\ m_2 \end{bmatrix}$ is then a bivariate t-distribution with 22 degrees of freedom with estimated mean.

$$\hat{\theta}_{11} = \begin{bmatrix} 1,293 \\ 2,478 \end{bmatrix} \cdot 10^{-3}$$

and estimated covariance matrix (from 2.12):

$$\hat{\Sigma} = \begin{bmatrix} 1,770 & 0,015 \\ 0,015 & 1,632 \end{bmatrix} \cdot 10^{-8}$$

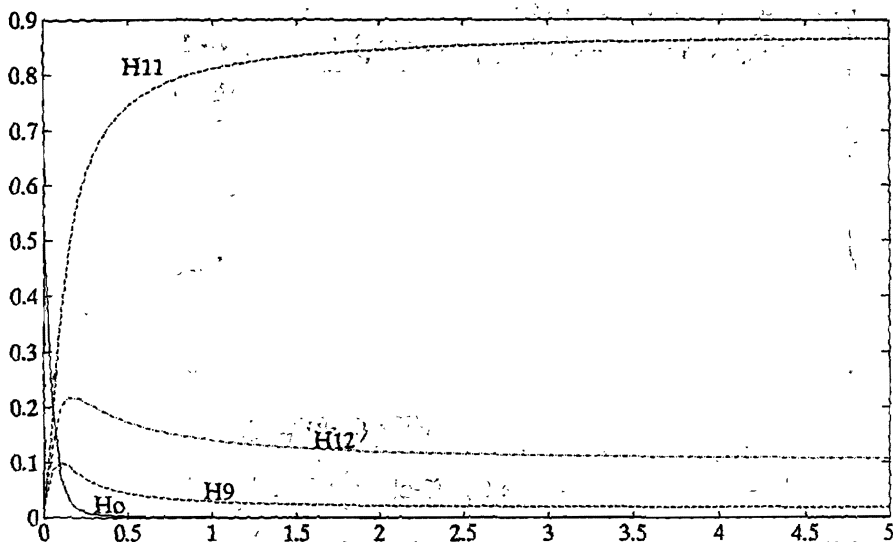


Figure 2. Posterior change-point probabilities

Here we have assumed, as did Simpkin and Downham, that there can only be one change-point. The analysis can easily be extended to provide for the possibility of more change-points.

Simpkin and Downham (1988) used the asymptotic distribution of the likelihood-ratio statistics to test the null hypothesis and it was rejected at

the 1 per cent level. They then examined the power of the test by means of simulations, assuming the change-point is at year-11.

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